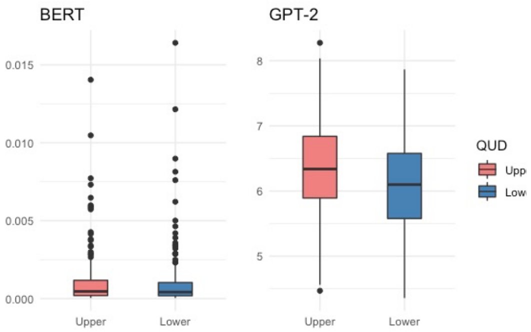

Figure 5. Distribution of surprisal scores for processing *some*-sentences across QUDs

text sequence. After providing this combined sequence as input to GPT-2, we analyzed the probability of the next generated token. This probability was used to estimate the likelihood of ANSWER appearing after QUESTION.

The output probabilities (*P*) from both models were transformed into **Surprisal** (Hale, 2001; Levy, 2008). Surprisal plays an effective role in measuring cognitive effort in human language processing. In this case, surprisal was used to measure models' processing difficulties. As shown in (10), this value is inversely correlated with how acceptable the next sentence (*S*) is in the given context (*Context*).

(10) $Surprisal = -log_2 P\ (S|Context)$

With surprisal scores, we could compare the processing difficulties of the models in the upper- and lower-bound QUDs. The overview of the experiment 1 is shown in Figure 4.

5.2 Result

Figure 5 depicted the distribution of surprisal scores for each model across QUDs. BERT showed little difference in surprisals based on QUDs (median of Upper = 0.00045, median of Lower = 0.00041), and statistically, no main effects were observed ($p = 0.48$). This suggested that BERT was unaffected by context in the interpretation of scalar implicature. Conversely, GPT-2 exhibited higher surprisal scores for the upper-bound QUD (median = 6.33) compared to the lower-bound QUD (median = 6.09), and this result was statistically significant ($p < 0.01$). This processing pattern of GPT-2 was consistent with human language processing, which suggested that GPT-2 showed processing difficulties, similar to the greater cognitive effort that humans expend in inferring scalar implicature in the context of QUD.

In summary, while exploring the interpretation of scalar implicature across QUDs, BERT exhibited no sensitivity to context, whereas GPT-2 clearly manifested the effects of context.

6 Discussion

Through two sets of experiments, this study investigated how LLMs interpret scalar implicature, between semantic entailment and pragmatic implicature, in the absence of context and how QUD, as a contextual cue, affects LLMs' processing of scalar implicature. Experiment 1 investigated whether *some*-sentences in BERT and GPT-2 exhibit greater similarity to semantic or pragmatic interpretations. The results showed that both models preferred the interpretation of pragmatic implicature over semantic entailment for *some*-sentences. Experiment 2 aimed to investigate whether providing QUD as a contextual cue would impact processing difficulties for BERT and GPT-2, comparing between the upper-bound QUD, where pragmatic implicature is required, and the lower-bound QUD, where implicature is not required. As a result, BERT showed no significant difference in processing difficulties based on QUDs, whereas GPT-2 showed more processing difficulties in the upper-bound QUD. In conclusion, this study found that, only in a certain language model, GPT-2, greater processing difficulties were captured during pragmatic inference of scalar implicature, aligning with human language processing.

BERT and GPT-2, despite both being built on the transformer architecture, exhibited markedly different patterns regarding their theoretical approaches to the processing of scalar implicature. Although both models shared the patterns of interpreting the term *some* as a pragmatic *not all* implicature rather than a semantic *at least one and possibly all* without context, BERT exhibited no discernible difference in processing based on QUDs. This can be explained by Default model where the meaning of *some* inherently defaults to *not all* (Levinson, 2000). On the other hand, GPT-2 represented a clear difference in processing difficulties when manipulating context through the setting of QUD, revealing that greater processing difficulties were captured in the processing of scalar implicature. This finding follows Context-driven model, consistent with the argument that *not* *all* implicature is not inherently embedded to the term *some* but rather inferred through a broader context (Wilson and Sperber, 1995; Sperber and Wilson, 2002).

Among the earlier NLI studies regarding scalar implicature, Jeretic et al. (2020) found that BERT learned scalar implicature. They claimed that positive results on scalar implicature inference, triggered by specific lexical items like *some* and *all*, probably exploits prior knowledge during the pre-training stage. The natural language data employed in the pre-training inherently include pragmatic information, which raises the possibility that such pre-training induces patterns of pragmatic inference in the data. Therefore, the results of the experiment 1 in this study, where the interpretation of pragmatic implicature occurred even in the absence of context, can be explained as leveraging inherent pragmatic information in the pre-training data of LLMs.

In the study of Schuster et al. (2020), which investigated the effects of linguistic features on scalar implicature, they found that their model could make accurate predictions without considering the preceding context, while incorporating the preceding conversational context did not enhance and even diminished prediction accuracy. This led to the assumptions that only a context-independent utterance is sufficient and contextual cues may not be necessary for pragmatic inference, or that the model has not appropriately used contextual information. Finding that context is unnecessary in scalar implicature may provide an explanation for our observation that BERT in the experiment 2 showed no difference in processing efforts across QUDs. However, this explanation may not generalize to effectively capture the processing of scalar implicature in all LLMs, especially when taking into account the effects of QUD on the processing of scalar implicature in GPT-2.

Liu et al. (2019) reported that features generated by pre-trained contextualizers were sufficient for achieving high performance across a broad range of tasks which explored the linguistic knowledge and transferability of contextualized word representations. However, they proposed that, for tasks requiring specific information not captured by contextual word representations, learning task-specific contextual features plays a crucial role in encoding the requisite knowledge. Within this framework, pragmatic implicature may either be pre-trained or require additional learning processes, depending on LLMs. Therefore, it is crucial to recognize that different language models may incorporate diverse linguistic information and exhibit distinct processing patterns for the same linguistic phenomenon.

Furthermore, based on the argument of Degen and Tanenhaus (2015, 2016) in which humans are influenced by context-driven expectations about unspoken alternatives, Hu et al. (2023) examined the BERT model's variation in scalar implicature rate not just within a single scale like <*some, all*> but also across scales with diverse lexical items as unspoken alternatives of *some.* This study revealed that the model's ability to make pragmatic inferences becomes stronger as more alternatives become available, which is depending on contextual predictability. This result leads us to expect that BERT will show contrasting result if more alternatives are presented and the context becomes more predictable, despite the failure to make pragmatic inference within the provided context in the present study.

In conclusion, the findings of this study suggested that LLMs are capable of pragmatic inference for scalar implicature without context. However, it is essential to understand the degree of contextual information utilization in each model and ensure appropriate learning for specific tasks.

7 Limitations

While this study has advanced our comprehension of pragmatic inference in LLMs regarding scalar implicature, it faces limitations in three aspects.

The first limitation is the absence of diverse constructions in which the scalar quantifier *some* appears. The exclusive use of experimental sentences featuring '*some* + NP' in the subject position within a single clause may not fully capture the broad spectrum of pragmatic interpretations that arise in various linguistic constructions and meanings in the real world. Additionally, the number of data used in the experiments might be not large enough to generalize the findings.

Secondly, the study relies on only two of early transformer-based models, which may not reflect the performance of more advanced models that have emerged recently. Since newer and more advanced models have demonstrated significantly higher performance across a wide range of tasks, using different models could yield varying results.

Lastly, in order to draw comparisons with human language processing, the experimental designs in this study deviate from conventional Natural Language Inference (NLI) tasks. Moreover, the metrics used in this study (i.e., cosine similarity or next sentence prediction) may yield different results when other metrics are applied. The diversity in experimental methodologies can lead to variations in results, emphasizing the necessity for future research to take into account such differences.

8 Conclusion

In this study, we discovered that LLMs interpret scalar implicature through pragmatic rather than semantic interpretation. Additionally, the study identified the model that engage in pragmatic inference through the processing of scalar implicature using a contextual cue, such as QUD, in contrast to the model that do not employ pragmatic inference. This study not only contributes to our comprehension of how LLMs process complex linguistic phenomena but also underscores the importance of considering pragmatics in NLP. By shedding light on the interplay between context and pragmatic inference, this study advances our understanding of LLMs and provides valuable insights for refining language models and applications in NLP.

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